

# Distinguishing novelty from contamination in hard-to-classify eDNA reads using a genomic language model's perplexity

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## Abstract

In eDNA metabarcoding a large fraction of reads return without a confident taxonomic label, and this "unclassified" pile mixes two biologically opposite cases that similarity-based assignment cannot tell apart: genuinely novel sequences worth investigating, and contaminated reads of known species that should be discarded. We ask whether the perplexity of a genomic language model — Evo2 (7B), used as a frozen feature source with no fine-tuning — supplies the missing axis. From 159 non-redundant arthropod COI reference barcodes we built a controlled dataset of 56,127 simulated 312-bp reads with exactly known labels: a novelty family evolved to a graded series of taxonomic distances (Pyvolve), and five contamination modes (Illumina error, PCR chimera, off-target insert, post-mortem damage, and a stacked control). Each read's per-position negative log-likelihood profile was scored once through Evo2, and small classification heads (scalar threshold, logistic, MLP, CNN) were compared under repeated, leakage-controlled cluster-atomic cross-validation. Detection rises monotonically with divergence and is near-perfect from the Family tier outward ( $\text{AUROC} \geq 0.96$ ). At the hard "near-novel" ranks the single perplexity scalar collapses to chance (Species  $\text{AUROC} \approx 0.49$ ), but the full positional profile read by a small nonlinear head recovers real signal (Species  $\approx 0.67$ , Genus  $\approx 0.88$ ); added depth and a convolutional prior do not improve further, localizing the residual limit to the scalar likelihood feature rather than the classifier. Novelty's signature is a diffuse, sequence-wide elevation, and near-novel reads are diffusely confused with low-divergence contamination rather than with any single artefact. Evo2 perplexity thus provides a usable novelty-versus-contamination axis across most of the gradient, with richer Evo2 embeddings the natural route to closing the near-novel gap.

## 1 · Introduction / Question

Environmental DNA (eDNA) metabarcoding has become a standard tool for surveying biodiversity: rather than identifying organisms one at a time, it amplifies and sequences a short marker gene from a bulk environmental sample — for arthropods, a ~312-bp window of the mitochondrial cytochrome c oxidase I (COI) barcode — and assigns each resulting read to a taxon by comparison against a reference database. A single run returns hundreds of thousands to millions of reads, so assignment is fully automated. In practice a substantial fraction of reads return without a confident taxonomic label, and this "unclassified" fraction is precisely where the most consequential cases hide: it holds both the genuinely new biology a survey hopes to discover and the quality problems it must discard.

A hard-to-classify read is ambiguous in two biologically opposite ways. It may represent a genuinely novel sequence — an organism whose reference is absent from the database — or it may be a contaminated read of an otherwise known species. These two cases call for opposite downstream actions: the first is a candidate discovery to investigate, the second an artefact to filter. Yet the assignment pipeline reports both simply as "unclassified."

The reason it cannot resolve the ambiguity is structural: standard taxonomic assignment is similarity-based — a best-hit identity above a threshold, or a naive-Bayes classifier's confidence — so both causes manifest identically, as the absence of a confident match. The pipeline never asks why a read is unlikely under the reference set, and so offers no axis along which novelty and contamination could be separated.

This work asks whether a genomic language model can supply that missing axis. Evo2 is a large autoregressive model of genomes: like a language model over text, it assigns each position a probability given the preceding context, and the negative log-likelihood (NLL) of the observed base measures how surprised the model is to see it. Aggregated over a read, this surprise is the read's perplexity. The intuition motivating the study is that the two causes of an unclassifiable read should leave differently shaped surprise — a genuinely novel sequence, diverged from everything the model has internalized along its whole length, as a diffuse, sequence-wide elevation in NLL; a contaminated read of a known species, mostly ordinary, as surprise concentrated where the artefact sits (the chimeric junction, the off-target insert, the damaged read end). Throughout, Evo2 is used only as a frozen feature source, with small classification heads trained on its outputs (Section 2.2).

Two lines of work meet in this question. The first is the long-standing problem of taxonomic assignment in metabarcoding, where reads falling below a reference-similarity threshold are routinely set aside as unassigned, with no further discrimination of the cause. The second is the rapid emergence of genomic language models together with the broader observation, in both natural-language and biological-sequence modelling, that a model's

likelihood is a useful signal for how typical or atypical a sequence is. A single perplexity scalar, however, summarizes an entire read and discards where the surprise falls — exactly the positional information the hypothesis below relies on.

Stated precisely, the central hypothesis is that novelty and contamination leave different signatures in a read's per-position NLL profile — novelty as a diffuse, sequence-wide elevation, contamination as a more localized disturbance — and that this difference is learnable from the profile even in the regime where a single scalar summary of it is not enough to separate the classes.

Concretely, the report addresses five questions:

- Q1. Does Evo2 perplexity carry any signal separating novelty from contamination — does the scalar mean-NLL beat chance?
- Q2. Does the per-position profile carry information the scalar summary discards?
- Q3. How does detectability change along the taxonomic-divergence gradient (Species → Phylum), and where does the hard regime lie?
- Q4. Does added model capacity (linear → MLP → CNN) raise the detection ceiling, or is the limit set by the feature rather than the classification head?
- Q5. When novelty is missed, what is it confused with?

## 2 · Methods

### 2.1 Dataset and labels

To study the question under controlled labels, a dataset was built in which the true status of every read — novelty or contamination, and which kind — is known exactly, because each read is simulated rather than collected from a real sample. From a curated set of 159 non-redundant arthropod COI reference barcodes (derived from MIDORI and deliberately de-duplicated to remove near-identical sequences), 56,127 reads of 312 bp were generated, split into a novelty family and a contamination family.

Novelty family (cohort C6). Novel sequences were produced with Pyvolve, a library that simulates molecular sequence evolution: starting from a real sequence, it applies random nucleotide substitutions along a phylogenetic branch under a chosen substitution model, so the output is a sequence that has diverged from the original by a controlled amount. Each of the 159 references was evolved in this way to a series of increasing taxonomic distances, set by the simulated branch length (substitutions per site): 0.01 for the Species tier, 0.04 for Genus, 0.07 for Family, 0.10 for Order and 0.15 for Phylum, plus two control anchors at 0.20 (Novelty) and 0.25 (Noise).

Two arthropod-specific precautions keep the simulated sequences biologically faithful and prevent the model from detecting the

simulation method instead of true novelty. First, the substitution model uses empirical base frequencies computed from the arthropod reference set, so the simulated reads match the nucleotide composition characteristic of arthropod COI. Second, each read is simulated over the exact same 312-bp window, reading frame and coordinate system as the contamination cohorts, so the novelty and contamination families are perfectly aligned and no coordinate offset can act as a shortcut the model might learn instead of the biology. Each tier was produced as 10 reads per reference (1,590 per tier), in matched error-free ("clean") and Illumina-error versions; the clean versions and all reference anchors are excluded from every train and test split, so the classifier cannot exploit the trivial cue that simulated reads are otherwise error-free.

Contamination family. Each reference was contaminated 30 times per mode (4,770 reads per cohort), modelling the five ways a read of a known species becomes hard to classify:

- C1 — Illumina error: a position-dependent sequencing-error model applies base substitutions whose rate rises toward the 3' end of the read.
- C2 — PCR chimera: two reference parents are joined at a random breakpoint (80–230 bp), as when two amplicons fuse during PCR.
- C3 — off-target insert: a 60–120 bp non-arthropod bacterial 16S fragment replaces a middle window of the read (donor GC content varied), modelling a co-amplified contaminant.
- C4 — Briggs damage (low / mid / high): the Briggs post-mortem deamination model (C→T at 5' ends, G→A at 3' ends) at three severities, plus Illumina error, modelling degraded eDNA.
- C5 — stacked: Briggs-mid + chimera + Illumina combined, a worst-case "kitchen-sink" control.

Together the seven contamination cohorts form a single shared negative pool of 33,390 reads. For the per-rank detection task (Section 2.3) a rank's 1,590 positives face this whole pool, so the task is strongly imbalanced ( $\approx 1:21$ ) — a fact that drives the choice of metric and of model-selection criterion (Section 2.6).

## 2.2 Evo2 perplexity features

Every read was scored once through Evo2 (7 billion parameters), a genomic language model trained on roughly 9 trillion nucleotides, run on a GPU in per-base mode. Evo2 is treated as frozen: it is never fine-tuned and no gradient flows into it, so the whole pipeline is a transfer-learning setup in which Evo2 supplies fixed features and only small heads are trained.

For a read  $x_1 x_2 \dots x_N$ , the model returns at each position the probability it assigns to the observed base given all preceding bases,  $P(x_i | x_1 \dots x_{i-1})$ . The per-position negative log-likelihood is

$$NLL_i = -\ln P(x_i | x_1 \dots x_{i-1}),$$

giving a profile of  $N = 311$  values for a 312-bp read (one next-token loss per transition). High local NLL marks positions the model finds surprising. The scalar summaries are the mean NLL and the perplexity:

$$\text{mean NLL} = (1/N) \sum NLL_i, \text{perplexity} = \exp(\text{mean NLL}).$$

Two feature views are derived from the same profile: a scalar view (mean NLL), the simplest possible summary; and a full-profile view that keeps all 311 positions and therefore preserves the positional structure a scalar discards.

## 2.3 Detection tasks and the taxonomic gradient

The headline task is binary: for a given novelty tier, separate that tier's reads (positives) from the shared contamination pool (negatives). Running it across the gradient (Species → Phylum, plus the two controls) traces how detectability changes with divergence; by design, Species is the lowest-divergence tier. The "near-novel" headline combines the two lowest-divergence tiers (Species + Genus).

A complementary multi-class diagnostic replaces the binary label with the full set of cohorts, to ask not just whether novelty is detectable but what it is confused with.

## 2.4 Leakage control and repeated cross-validation

The dominant risk in this setting is that reads tracing to the same organism leak between training and test folds, rewarding memorization and inflating performance. This is controlled with cluster-atomic group k-fold cross-validation. Reads are grouped into clusters by their 312-bp source reference: every read built from a single reference — its C1, C3 and C4 contamination versions together with its C6 novelty reads — shares that reference's cluster. The two-parent cohorts (chimeras C2 and stacked C5) descend from two references at once and so cannot belong to a single cluster; they are governed by the boundary rule below. Each cluster is then assigned whole to a single fold ("atomic": it is never split across folds), so the model is never tested on an organism whose other reads it saw in training.

This boundary rule affects only the two-parent cohorts — chimeras (C2) and stacked reads (C5), each of which fuses two references — and only the reads whose two parents happen to fall in different folds. Such a read straddles the train/test line: it cannot be placed in either fold without dragging one parent's sequence across the boundary, so it is flagged boundary and dropped automatically as the folds are formed. The single-parent cohorts (C1, C3, C4) are never affected and are retained in full; for the two-parent cohorts, roughly four in five reads are dropped in a given partition ( $\approx 919$  of 4,770 C2 reads and  $\approx 902$  of 4,770 C5 reads are kept). Crucially, a read is dropped for a reason unrelated to its biology — purely which two references co-locate — so the retained chimera and stacked reads remain a representative, unbiased subset that the model still trains on and is tested against, only at reduced sample size. The contamination negatives therefore remain well populated ( $\approx 25,700$  reads, dominated by the fully-retained single-parent cohorts).

To obtain honest error bars rather than five correlated numbers, the entire partition is regenerated under five seeds (42, 101, 202, 303, 404), each a full, leak-free 5-fold CV; cluster atomicity is re-verified on every repeat. Every replicate is recorded individually and never pre-averaged: each (repeat  $\times$  fold) run — and, for the neural models, each initialization seed within it — contributes its own performance number, so the final error bar reflects the full spread across partition, fold, and initialization rather than hiding seed variance behind an early average.

Within each five-fold partition the folds play distinct roles. For the neural rungs, one fold is held out as the test set, a second — the cyclically adjacent fold — as a validation set used solely to choose the early-stopping epoch, and the remaining three folds form the training set; rotating the test fold through all five positions yields five (train / validation / test) splits per repeat. Because folds are cluster-atomic, the validation fold is leak-free with respect to both training and test, exactly as the test fold is. The scalar and logistic rungs make no early-stopping decision and so hold out no validation fold: each tests on one fold and trains on the other four. The neural rungs are therefore trained on slightly less data (three folds rather than four), so any capacity advantage they show is measured, if anything, conservatively.

Feature standardization — rescaling each of the 311 positions to mean 0 and standard deviation 1 — is fit on the training folds only and applied to validation and test, so no test statistic informs preprocessing.

## 2.5 The model-capacity ladder

Rather than tuning a single architecture, the models are compared as a deliberate ladder of increasing capacity under the identical CV protocol, so that any gain is attributable to capacity and not to evaluation differences. The rungs are: (i) the scalar mean-NLL threshold; (ii) a logistic regression on the full 311-position profile (class-weight balanced); (iii) a tiny MLP (311 → 64 → 1, ReLU, dropout 0.10); and (iv) a tiny 1-D CNN (Conv1d(1→8, k5) → ReLU → Conv1d(8→8, k5) → ReLU → global pool → Linear(8→1)), the only rung able to exploit positional structure; average- and max-pooling variants were both evaluated.

Because the experiment treats capacity as the controlled variable, each rung's size is stated explicitly below. The heads consume the per-position NLL profile (or, for the scalar rung, its mean); Evo2

**Table 1.** The model-capacity ladder: input features and trainable-parameter counts for each rung. Evo2 contributes ~7 billion frozen, untrained parameters and is excluded from the counts.

| Rung                               | Input features  | Trainable parameters     |
|------------------------------------|-----------------|--------------------------|
| (i) Scalar mean-NLL threshold      | 1 (mean NLL)    | 1 (a decision threshold) |
| (ii) Logistic regression (profile) | 311             | 312                      |
| (iii) Tiny MLP (311 → 64 → 1)      | 311             | 20,033                   |
| (iv) Tiny 1-D CNN                  | 1 channel × 311 | 385                      |

The first three rungs increase in capacity — from a single threshold, to a linear map over all 311 positions, to a nonlinear two-layer network of ~20k weights. The CNN is deliberately not a further step up in parameter count: its weight-shared, 5-wide filters give it only 385 trainable parameters, far fewer than the MLP's. Its place in the comparison is a change in inductive bias — local, translation-equivariant filters — rather than an increase in raw capacity, and whether that spatial prior helps is the architecture-side question taken up at the hard ranks in Section 3.4.

Set against the ~16,000 reads in a training partition — of which only ~950 are positives — the MLP's ~20k parameters place it just past the point where parameters exceed training samples, the regime in which overfitting is expected; this is precisely why dropout and early stopping (below) are part of its recipe, and why the much smaller logistic (312) and CNN (385) heads, sitting well inside the data-rich regime, instead tend toward underfitting. The input dimensionality is modest by comparison (311 features ≪ 16,000 samples), so the risk lies on the parameter side, not in the feature count.

The neural rungs share one recipe: Adam (learning rate 1e-3), BCEWithLogitsLoss, at most 200 epochs, with early stopping (patience 25) restoring the best-epoch weights. The stopping epoch is chosen on a validation metric — never on the test fold, whose single held-out role would otherwise be spent. For the MLP that metric is the validation average precision: under the 1:21 imbalance, AP is far more sensitive than AUROC to the one improvement that matters while training — lifting the rare positives toward the top of the ranking — and so makes the better steering signal, even though AUROC (prevalence-independent, and therefore comparable across ranks of differing prevalence) remains the headline metric reported in Section 2.6. For the CNN, whose validation AP is unstable given the few positives per fold, the minimum-validation-loss epoch is used instead. Each (repeat × fold) is trained from five initialization seeds so init variance is visible rather than hidden. The models are tiny and their hyperparameters were fixed by design rather than searched, to avoid overfitting the small, group-atomic dataset.

## 2.6 Evaluation

Because the task is imbalanced, accuracy is not used as a headline metric. AUROC is reported as the primary measure — it is threshold- and prevalence-independent, which makes it comparable across tiers of differing prevalence — alongside average precision (AP), the area under the precision-recall curve, which captures behaviour where positives are rare. Error bars are the mean ± standard deviation across all replicates (repeat × fold, and × seed for the neural models).

itself contributes ~7 billion parameters as a frozen backbone — none of them trained — so the counts below are the trainable heads alone:

For the multi-class diagnostic, the softmax classifier is trained with a class-weighted ("balanced") loss so the dominant cohort cannot drive training, and the confusion matrix is summed over folds (and seeds) and then row-normalized (rows = true class). Reading each row as that true class's own distribution over predictions makes every class contribute equally regardless of size, so a large cohort cannot visually swamp a small one.

## 2.7 Implementation and reproducibility

Evo2 per-base scoring and the full multi-seed neural sweep were run on a rented GPU (Lightning AI); baselines, the deterministic ladder, figures, and diagnostics were produced on a MacBook Air M4. All randomness is seeded (partition, fold, and initialization seeds are recorded), and a reproducibility log records library versions and the exact commands. The analysis is driven by a single engine module (`master_paper_lib.py`) and a master notebook, so the dataset can be swapped from one source of truth without code drift.

## 3 • Results

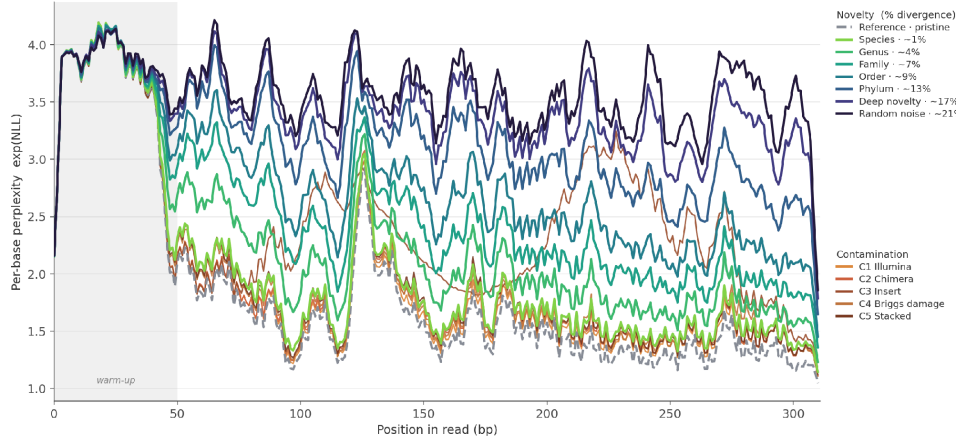
### 3.1 Per-position perplexity shows a clear novelty gradient; contamination mostly overlaps it (Fig 1)

Before any classifier, the per-position perplexity profiles already show one half of the hypothesized picture clearly (Fig 1). The novelty cohorts form an ordered fan: a broad, sequence-wide lift in perplexity that grows monotonically with divergence, from the Species tier just above the contamination band up to the Phylum and noise anchors at the top, with no single position responsible. This diffuse, whole-read elevation is exactly the novelty signature the hypothesis predicts.

The contamination side is only partly visible in this averaged view. All contamination cohorts except the off-target insert (C3) sit low and close together, in the band between the Species and Genus novelty curves, and overlap one another — so the mean per-base profile does not let us pinpoint where their surprise departs from low-divergence novelty's. The lone clear exception is C3, whose raised block over the inserted window stands out as a distinct, localized signature. In other words, Fig 1 confirms the diffuse-novelty half of the hypothesis and the localized signature of the insert directly; the localized signatures expected for the other contamination modes are not resolvable from these averaged curves, and the case that the profile still carries usable information at low divergence is made by the classifiers below rather than by eye here.

Fig 1 · novelty gradient vs contamination · COI · 311 bp

**Per-base perplexity rises with evolutionary divergence; contamination stays low**



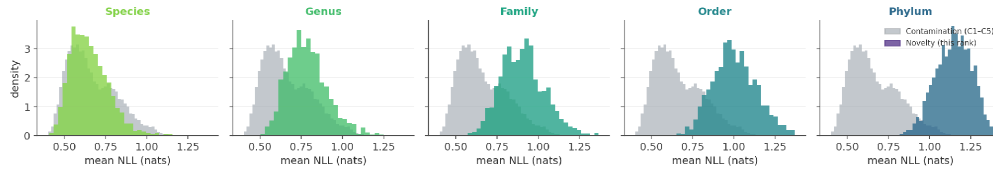
**Figure 1.** Per-base perplexity along the read. Novelty cohorts form an ordered, divergence-scaled fan; contamination cohorts (except the off-target insert, C3) sit low and overlapping between the Species and Genus novelty curves.

### 3.2 A scalar perplexity visibly overlaps at low divergence (Fig 2)

Collapsing each profile to its mean NLL — a single perplexity scalar — and plotting the novelty and contamination distributions rank by rank shows directly where the scalar runs out (Fig 2). At Species the novel reads' mean-NLL histogram lies almost entirely on top of the contamination pool: the two distributions overlap nearly completely, so no threshold on the scalar can cleanly separate them. As divergence grows the novelty distribution shifts toward higher mean NLL and pulls away from the (fixed) contamination pool — still

substantially overlapping at Genus, then progressively cleaner at Family, Order and Phylum, where the two distributions are largely distinct. The visual message is that a single scalar is only sufficient once divergence is large, and fails by inspection at the lowest-divergence ranks — which is what motivates keeping the full positional profile and moving beyond the scalar. (How much each model actually recovers from the profile is quantified by AUROC in §3.3.)

Fig 2 · Scalar mean-NLL: novelty and contamination overlap at low divergence



**Figure 2.** Distribution of the scalar mean-NLL for novelty (coloured) versus the shared contamination pool (grey) at each rank. The two distributions overlap almost completely at Species and separate progressively toward Phylum.

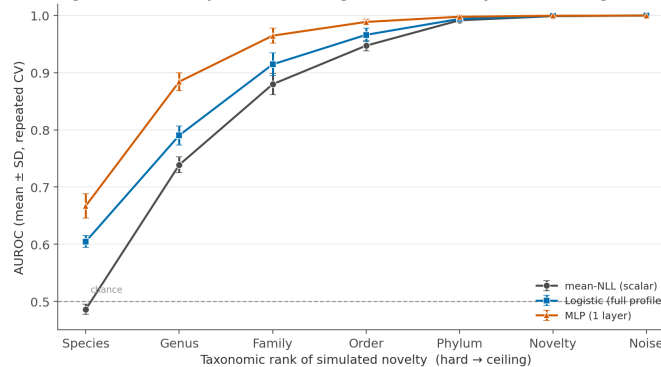
### 3.3 Detectability rises monotonically along the taxonomic gradient (Fig 3)

Quantifying separation as AUROC and running it across the gradient (Fig 3, Q3) turns the visual overlap of §3.2 into numbers. Three things are read directly off the figure. First, every model's curve rises monotonically with divergence: separation improves at every step from Species to Phylum and saturates at the two control anchors. Second, the three models are cleanly ordered at every rank — the MLP curve sits above the logistic curve, which sits above the scalar — so using the full 311-position profile beats the scalar, and a small nonlinearity beats a linear read of the same profile (this ordering is

taken up in §3.4). Third, detection becomes near-perfect from the Family tier outward, where all three curves crowd together near the top and the error bars shrink.

The difficulty is therefore concentrated entirely in the two lowest-divergence tiers, Species and Genus — the "near-novel" regime, where a new organism sits within a few percent identity of a known reference. Here the gap between models is widest and the scalar collapses to the chance line at Species (Species AUROC  $\approx$  0.49) while clearing chance comfortably at every other rank (Q1), and the MLP retains a clear margin above it. The remaining analysis focuses on these two ranks.

Fig 3 · Detection improves with divergence; near-novelty is the hard regime



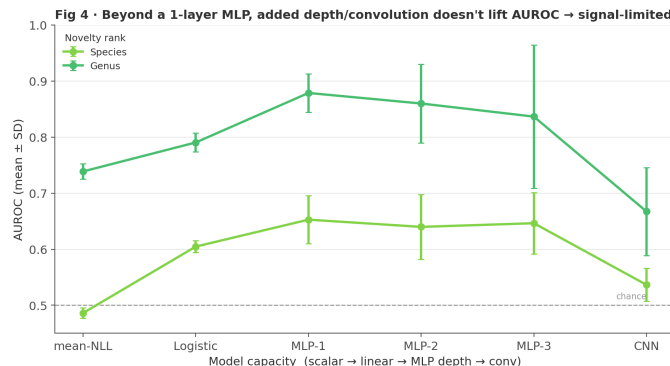
**Figure 3.** Per-rank detection AUROC (mean  $\pm$  SD, repeated CV) for the scalar, logistic and one-layer MLP heads. Detectability rises monotonically with divergence and is near-perfect from the Family tier outward; the hard regime is Species–Genus.



### 3.4 A small nonlinear head sets the hard-rank ceiling — and capacity then plateaus (Fig 4)

Fig 4 walks the two hard ranks (Species, Genus) along a capacity axis — scalar mean-NLL → logistic profile → a one-layer MLP, then deeper MLPs (two and three layers) and finally the 1-D CNN — with AUROC (mean  $\pm$  SD) on the y-axis. Two findings are visible.

First, the curve climbs over its first three rungs: from the scalar, to the linear profile model, to the one-layer MLP, which is the high point of the figure at both ranks. This is the §3.3 ordering seen as a ladder — the profile's positional structure carries information a single scalar discards, and a small nonlinearity extracts more of it than a linear model can (Q2).



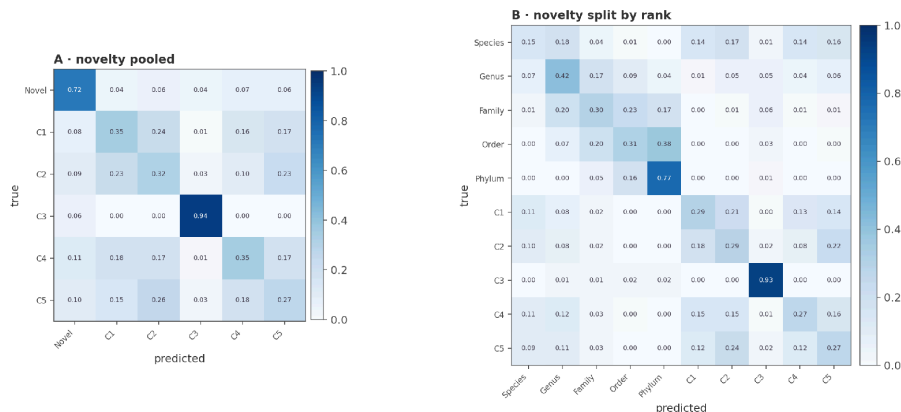
**Figure 4.** Capacity ladder at the two hard ranks (Species, Genus). AUROC climbs from the scalar to the one-layer MLP, then plateaus across deeper MLPs and falls for the CNN — the ceiling is set by the feature, not the head.

### 3.5 Near-novelty is diffusely confused with low-divergence contamination — except inserts (Fig 5)

The multi-class diagnostic replaces the binary label with the full set of cohorts and asks what near-novel reads are actually mistaken for (Q5). In the balanced, row-normalized confusion matrix (Fig 5), a Species-tier novel read is recognized as novelty (predicted Species or Genus) only about a third of the time; the rest of its probability mass is spread almost evenly across the contamination cohorts — C1  $\approx$  14 %, C2  $\approx$  17 %, C4  $\approx$  14 %, C5  $\approx$  16 % — rather than concentrated on

any single one. The lone exception is the off-target insert C3, which absorbs only  $\approx$  1 % of novel reads and is itself correctly recognized  $\approx$  93 % of the time: its localized, compositionally distinct block is easy to separate from diffuse novelty. Near-novelty is thus not mistaken for one specific artefact but is diffusely confused with the whole family of low-divergence contamination — consistent with the scalar's failure at Species (§3.2) and with novelty's signature being a non-specific, sequence-wide elevation rather than a localized event.

**Fig 5 • Novelty is recovered overall (A), but low-divergence novelty is diffusely confused with low-divergence contamination (B)**



**Figure 5.** Balanced, row-normalised confusion matrices. (A) novelty pooled; (B) novelty split by rank. Near-novel reads are diffusely confused with the whole family of low-divergence contamination, the lone exception being the off-target insert C3.

## 4 • Discussion

The central result is that a frozen genomic language model supplies the axis that similarity-based assignment lacks. Evo2's perplexity — computed without any fine-tuning, as a pure transfer-learning feature — carries a real, learnable signal that separates novelty from contamination, and it does so in the way the hypothesis predicted. Across the divergence gradient the separation is not merely present but well-behaved: it rises monotonically with divergence (Fig 3) and becomes near-perfect from the Family tier outward (MLP AUROC  $\geq$  0.96), saturating on the control anchors. For the large part of the

gradient where a novel organism is more than a few percent diverged from anything in the reference set, a small head on Evo2 perplexity already turns the undifferentiated "unclassified" bucket into a confident novelty-versus-artefact call.

The mechanism behind that signal is the one the study set out to test. The per-position profiles (Fig 1) show novelty as a diffuse, sequence-wide elevation in perplexity that grows with divergence, exactly the signature expected of a sequence that has drifted away from everything the model has internalized along its whole length. The multi-class diagnostic (Fig 5) reinforces this from the opposite direction: when a near-novel read is misread, its probability mass is

not captured by any single contamination mode but spread almost evenly across the low-divergence artefacts — a diffuse confusion that is itself the fingerprint of a non-specific, whole-read signal rather than a localized event. The one contamination mode with a sharply localized, compositionally distinct signature, the off-target insert (C3), is both visibly separable in the raw profiles and almost never confused with novelty (recognized  $\approx 93\%$  of the time). The picture is therefore internally consistent: localized artefacts that look unlike diffuse novelty are easy to set apart, and the difficulty is concentrated precisely where novelty's diffuse signature is weakest — at the lowest divergence.

That low-divergence regime is where the deep-learning contribution of the work is clearest. A single perplexity scalar — the mean NLL — collapses to chance at the Species tier (Fig 2), where the novel and contamination distributions overlap almost completely. Keeping the full 311-position profile and reading it with a linear model recovers part of the signal the scalar discards, and a small nonlinear head recovers more still, lifting Species AUROC from chance to  $\approx 0.67$  and Genus to  $\approx 0.88$  (Fig 4). This is a concrete demonstration of the report's second question: positional structure that a scalar summary throws away is real and is extractable, but only by a model that looks at the profile as a whole rather than at its average.

It is equally informative that the ladder stops climbing there. Added MLP depth does not raise AUROC, and a convolutional head with an explicit spatial prior does not exceed the one-layer MLP (Fig 4). When more capacity and a different inductive bias both fail to help, the limit is not the classifier — it is the feature. This is a useful negative to have localized rather than a disappointment: it says the residual difficulty at near-novelty lives in Evo2's scalar likelihood, a single compressed number per position, and not in the size or shape of the head trained on it. The likelihood is the most lossy view the model can offer; the natural way to raise the ceiling is to give the head a richer view of the same frozen backbone.

Read back against the motivating problem, the results draw a clear line through the "unclassified" fraction. Wherever a query is more than roughly a family's worth of divergence from its nearest reference, Evo2 perplexity supports a confident novelty call and a contaminated read of a known species can be told apart from a genuine discovery. In the near-novel corner — a new organism within a few percent identity of a known barcode — the perplexity scalar alone cannot make that distinction, and even the profile leaves the two partly entangled, so a practical pipeline would be right to treat near-novel candidates and low-divergence contamination as a single "needs review" tier rather than auto-sorting them. These conclusions rest on a deliberately controlled setting — novelty simulated with Pyvolve and contamination synthesized cohort by cohort, a single COI marker in arthropods, and Evo2 used only as a frozen scalar-likelihood source — which is what makes the labels exact, but also what the next steps must relax.

Three follow on directly. First, and most pointed by the feature-limited ceiling, is to replace the scalar likelihood with Evo2's internal embeddings: the same frozen model exposes a high-dimensional representation at each position, with far more headroom than one likelihood number, and the architecture-side result predicts that this — not a bigger head — is where the near-novel gain should come from. Second is validation on real eDNA reads, where true novelty and real contamination replace their simulated stand-ins and the diffuse-versus-localized account can be tested outside the generative model that produced it. Third, the positional profile invites moving from detection to localization — using where the surprise concentrates to point at the contaminating junction or insert directly, turning the same signal that flags an artefact into an explanation of it. Together these keep the frozen-backbone, small-head design that makes the approach cheap and reproducible, while aiming the next round of work at the one regime this study leaves open.

## 5 • Conclusion

This work asked whether a genomic language model's perplexity can separate genuine novelty from contamination among the eDNA reads that similarity-based assignment leaves unclassified, and posed five specific questions; the answers can now be stated directly. Evo2 perplexity does carry a real separating signal (Q1): as a single

mean-NLL scalar it clears chance comfortably across most of the divergence gradient, failing only at the lowest-divergence Species tier, where the novel and contamination distributions overlap almost completely. That failure is not the end of the signal, because the per-position profile carries information the scalar discards (Q2): read by a small nonlinear head it lifts the hardest rank from chance to AUROC  $\approx 0.67$ , and Genus to  $\approx 0.88$ . Detectability rises monotonically along the gradient and becomes near-perfect from the Family tier outward, so the difficulty is confined to the "near-novel" regime within a few percent identity of a known reference (Q3). Added capacity does not move that ceiling — neither MLP depth nor a convolutional spatial prior beats the one-layer MLP — which locates the residual limit in Evo2's scalar-likelihood feature rather than in the classification head (Q4). And when near-novelty is missed, it is mistaken not for any single artefact but diffusely for the whole family of low-divergence contamination (Q5), exactly the fingerprint expected of a non-specific, sequence-wide novelty signature. Taken together, Evo2 perplexity supplies a usable novelty-versus-contamination axis across most of the gradient and points to its own next step: richer Evo2 embeddings, in place of the scalar likelihood, as the route to closing the near-novel gap.